

ALFAAL03000

## N,N-Dimethylformamide

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | N,N-Dimethylformamide<br>N,N-Dimethylformamide                                                                                                                                                                                                                     |
| Cat No. :                     | L03000                                                                                                                                                                                                                                                             |
| Synonyms                      | DMF                                                                                                                                                                                                                                                                |
| CAS No                        | 68-12-2                                                                                                                                                                                                                                                            |
| Molecular Formula             | C3 H7 N O                                                                                                                                                                                                                                                          |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                     |                         |                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------|
| Physical State<br>Liquid                                                                                                                                                                                            | Appearance<br>Colorless | Odor<br>Rotten-egg like |
| <b>Emergency Overview</b><br>May be harmful if swallowed. Flammable liquid and vapor. Causes serious eye irritation. May damage fertility or the unborn child.<br>Harmful in contact with skin. Harmful if inhaled. |                         |                         |

#### Classification of the substance or mixture

|                                    |             |
|------------------------------------|-------------|
| Flammable liquids.                 | Category 3  |
| Acute Oral Toxicity                | Category 5  |
| Acute Dermal Toxicity              | Category 4  |
| Acute Inhalation Toxicity - Vapors | Category 4  |
| Serious Eye Damage/Eye Irritation  | Category 2  |
| Reproductive Toxicity              | Category 1B |

#### Label Elements

**Signal Word****Danger****Hazard Statements**

H226 - Flammable liquid and vapor  
H303 - May be harmful if swallowed  
H319 - Causes serious eye irritation  
H312 + H332 - Harmful in contact with skin or if inhaled  
H360 - May damage fertility or the unborn child

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Vapors may cause flash fire or explosion. Flammable liquid.

**Health Hazards**

May be harmful if swallowed. Causes serious eye irritation. May damage fertility or the unborn child. Harmful in contact with skin. Harmful if inhaled.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility but will likely degrade over time. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component         | CAS No  | Weight % |
|-------------------|---------|----------|
| Dimethylformamide | 68-12-2 | >95      |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Inhalation**

Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention.

**Ingestion**

Do NOT induce vomiting. Get medical attention.

**Most important symptoms and effects**

Irritating to eyes. Difficulty in breathing. May be harmful if absorbed through skin: Gastrointestinal discomfort: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use water jetstream.

**Specific Hazards Arising from the Chemical**

Flammable. Risk of ignition. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

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## SECTION 7. HANDLING AND STORAGE

**Handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

**Specific Use(s)**

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control Parameters**

| Component         | China                             | Taiwan                                   | Thailand    | Hong Kong                                |
|-------------------|-----------------------------------|------------------------------------------|-------------|------------------------------------------|
| Dimethylformamide | TWA: 20 mg/m <sup>3</sup><br>Skin | TWA: 10 ppm<br>TWA: 30 mg/m <sup>3</sup> | TWA: 10 ppm | TWA: 10 ppm<br>TWA: 30 mg/m <sup>3</sup> |

  

| Component         | ACGIH TLV          | OSHA PEL                                                                                                         | NIOSH                                                     | The United Kingdom                                                                                                    | European Union                                                                                                                                                                      |
|-------------------|--------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimethylformamide | TWA: 5 ppm<br>Skin | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 30 mg/m <sup>3</sup><br>Skin<br>TWA: 10 ppm<br>TWA: 30 mg/m <sup>3</sup> | IDLH: 500 ppm<br>TWA: 10 ppm<br>TWA: 30 mg/m <sup>3</sup> | STEL: 10 ppm 15 min<br>STEL: 30 mg/m <sup>3</sup> 15 min<br>TWA: 5 ppm 8 hr<br>TWA: 15 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 15 mg/m <sup>3</sup> (8h)<br>TWA: 5 ppm (8h)<br>Skin<br><br>STEL: 10 ppm (15min)<br>STEL: 30 mg/m <sup>3</sup> (15min)<br>STEL: 30 mg/m <sup>3</sup> (8h)<br>STEL: 10 ppm (8h) |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments                           |
|----------------|-------------------|-----------------|-------------|------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.5 mm          | EN 374      | As tested under EN374-3 Determination of |
| Neoprene       | < 100 minutes     | 0.45 mm         |             | Resistance to Permeation by Chemicals    |

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Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                                                                                                                                                                                                    |
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Type A Organic gases and vapours filter Brown conforming to EN14387                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains.                                                                                                                                                                                                                                                                                       |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                               |                                         |
|------------------------------------------------|-----------------------------------------------|-----------------------------------------|
| <b>Appearance</b>                              | Colorless                                     |                                         |
| <b>Physical State</b>                          | Liquid                                        |                                         |
| <b>Odor</b>                                    | Rotten-egg like                               |                                         |
| <b>Odor Threshold</b>                          | No data available                             |                                         |
| <b>pH</b>                                      | 6-8 @ 20°C                                    | 20% aq.sol                              |
| <b>Melting Point/Range</b>                     | -61 °C / -77.8 °F                             |                                         |
| <b>Softening Point</b>                         | No data available                             |                                         |
| <b>Boiling Point/Range</b>                     | 153 °C / 307.4 °F                             |                                         |
| <b>Flash Point</b>                             | 58 °C / 136.4 °F                              | <b>Method -</b> Abel-Pensky (DIN 51755) |
| <b>Evaporation Rate</b>                        | 0.17                                          | (Butyl Acetate = 1.0)                   |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                  |
| <b>Explosion Limits</b>                        | <b>Lower</b> 2.2 vol%<br><b>Upper</b> 16 vol% |                                         |
| <b>Vapor Pressure</b>                          | 4.9 mbar @ 20 °C                              |                                         |
| <b>Vapor Density</b>                           | 2.5                                           | (Air = 1.0)                             |
| <b>Specific Gravity / Density</b>              | 0.945                                         | @ 20 °C                                 |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                  |
| <b>Water Solubility</b>                        | Soluble                                       |                                         |
| <b>Solubility in other solvents</b>            | No information available                      |                                         |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                         |
| <b>Component</b>                               | <b>log Pow</b>                                |                                         |
| Dimethylformamide                              | -1.028                                        |                                         |
| <b>Autoignition Temperature</b>                | 445 °C / 833 °F                               |                                         |
| <b>Decomposition Temperature</b>               | > 350°C                                       |                                         |
| <b>Viscosity</b>                               | 0.8 mPa.s at 20 °C                            |                                         |
| <b>Explosive Properties</b>                    | Not explosive                                 | explosive air/vapour mixtures possible  |
| <b>Oxidizing Properties</b>                    | No information available                      |                                         |

## N,N-Dimethylformamide

Molecular Formula C3 H7 N O  
Molecular Weight 73.09  
Surface tension 36.42 mN/m (25 °C)

## SECTION 10. STABILITY AND REACTIVITY

**Stability** Stable under normal conditions.

**Hazardous Reactions** None under normal processing.

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Conditions to Avoid** Incompatible products. Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition.

**Materials to avoid** Strong oxidizing agents. Halogens. Halogenated compounds. Reducing Agent. .

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component         | LD50 Oral          | LD50 Dermal                             | LC50 Inhalation     |
|-------------------|--------------------|-----------------------------------------|---------------------|
| Dimethylformamide | 3040 mg/kg ( Rat ) | 1500 mg/kg (Rabbit)<br>3.2 g/kg ( Rat ) | >5.58 mg/L/4h (Rat) |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2  
**Test species** rabbit  
**Observation end point** Irritating to eyes

## (d) respiratory or skin sensitization;

**Respiratory** Based on available data, the classification criteria are not met  
**Skin** Based on available data, the classification criteria are not met

| Component                            | Test method                            | Test species | Study result      |
|--------------------------------------|----------------------------------------|--------------|-------------------|
| Dimethylformamide<br>68-12-2 ( >95 ) | Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | - non-sensitising |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component         | EU | UK | Germany | IARC     |
|-------------------|----|----|---------|----------|
| Dimethylformamide |    |    |         | Group 2A |

(g) reproductive toxicity; Category 1B  
**Reproductive Effects** Experiments have shown reproductive toxicity effects on laboratory animals.  
**Developmental Effects** May cause harm to the unborn child. Developmental effects have occurred in experimental animals.  
**Teratogenicity** Teratogenic effects have occurred in experimental animals.

## SAFETY DATA SHEET

## N,N-Dimethylformamide

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed May be harmful if absorbed through skin: Gastrointestinal discomfort: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

| Component         | Freshwater Fish                                                                                                                          | Water Flea           | Freshwater Algae     | Microtox                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|-------------------------------------------------|
| Dimethylformamide | Pimephales promelas:<br>LC50 = 10.6 g/L/96h<br>Onchorhynchus mykiss:<br>LC50 = 9.8 g/L/96h<br>Lepomis macrochirus:<br>LC50 = 6.3 g/L/96h | EC50 = 7500 mg/L/48h | EC50 = 7500 mg/L/96h | EC50 = 2000 mg/L 5 min<br>EC50 = 570 mg/L 240 h |

Persistence and Degradability Readily biodegradable  
Persistence Persistence is unlikely.

| Component                            | Degradability           |
|--------------------------------------|-------------------------|
| Dimethylformamide<br>68-12-2 ( >95 ) | 100 % (OECD 301E (21d)) |

Degradation in sewage treatment plant Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

Bioaccumulative Potential Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Dimethylformamide | -1.028  | 0.3 - 1.2 L/kg                |

Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility but will likely degrade over time Will likely be mobile in the environment due to its water solubility Highly mobile in soils

Surface tension 36.42 mN/m (25 °C)

## Endocrine Disruptor Information

| Component         | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Dimethylformamide | Group III Chemical                       |                                                  |                                         |

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

## N,N-Dimethylformamide

## Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

UN-No UN2265  
Proper Shipping Name N,N-DIMETHYLFORMAMIDE  
Hazard Class 3  
Packing Group III

## IMDG/IMO

UN-No UN2265  
Proper Shipping Name N,N-DIMETHYLFORMAMIDE  
Hazard Class 3  
Packing Group III

## IATA

UN-No UN2265  
Proper Shipping Name N,N-DIMETHYLFORMAMIDE  
Hazard Class 3  
Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component         | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Dimethylformamide | X                                                   | X                                       | X    | X     | 200-679-5 | X    | X   | X     | X    | X    | X    | KE-11411 |

## National Regulations

| Component                            | Toxic Chemical Substances Control Act |
|--------------------------------------|---------------------------------------|
| Dimethylformamide<br>68-12-2 ( >95 ) | Class II (30 wt%)<br>TRQ = 50 kg      |

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Creation Date 03-Sep-2009  
Revision Date 26-Apr-2024



## N,N-Dimethylformamide

## Revision Summary

New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**